

BACKLOG BUDDY ? STUDY MATERIAL

Computer Organization and Architecture ? Unit-wise Lecture Notes

Subject Code: CS304 Semester: 3

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Condensed notes covering all five units: Basic Structure of Computers, Machine Instructions and Addressing, ALU and Data Path Design, Memory Hierarchy and Cache, Input/Output Organization.

Unit 1: Basic Structure of Computers

- Definition, key concepts and significance of Basic Structure of Computers
- Important diagrams / derivations / algorithms for Basic Structure of Computers
- Solved examples and previous-year questions on Basic Structure of Computers

Unit 2: Machine Instructions and Addressing

- Definition, key concepts and significance of Machine Instructions and Addressing
- Important diagrams / derivations / algorithms for Machine Instructions and Addressing
- Solved examples and previous-year questions on Machine Instructions and Addressing

Unit 3: ALU and Data Path Design

- Definition, key concepts and significance of ALU and Data Path Design
- Important diagrams / derivations / algorithms for ALU and Data Path Design
- Solved examples and previous-year questions on ALU and Data Path Design

Unit 4: Memory Hierarchy and Cache

- Definition, key concepts and significance of Memory Hierarchy and Cache
- Important diagrams / derivations / algorithms for Memory Hierarchy and Cache
- Solved examples and previous-year questions on Memory Hierarchy and Cache

Unit 5: Input/Output Organization

- Definition, key concepts and significance of Input/Output Organization
- Important diagrams / derivations / algorithms for Input/Output Organization
- Solved examples and previous-year questions on Input/Output Organization

Exam Tips:

- * Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- * Solve at least 3 previous year papers under timed conditions before the exam.
- * Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- * Revise definitions, diagrams and standard derivations ? they earn easy marks.
- * Practise numerical problems from every unit; papers usually include 2?3 compulsory

problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.